

Predicting Personality Traits: An Analytical Study on Introversion and Extroversion Using Behavioral Data

Introduction:

Understanding personality traits such as introversion and extroversion can play a crucial role in fields like psychology, human resources, education, and personalized marketing. This study focuses on classifying individuals as **Introverts (0)** or **Extroverts (1)** using machine learning techniques applied to behavioral data.

The dataset includes features such as **time spent alone**, **stage fear**, **social event attendance**, **frequency of going outside**, **mood after socializing**, **friend circle size**, and **social media posting frequency**. These features are analyzed for their distribution, correlation, and predictive power.

A **confusion matrix** evaluates the model's classification performance, indicating strong accuracy in distinguishing between personality types. The **correlation matrix** helps identify which features are strongly associated with each other and with the personality outcome. The **distribution plots** give insights into the nature of each feature, helping to understand how behaviors differ among personality types.

This analysis demonstrates the effectiveness of behavioral indicators in predicting personality and provides a foundation for more personalized and human-centric applications in AI and data science.

Objectives:

1. **To classify individuals as Introverts or Extroverts** based on behavioral and social activity-related data.
2. **To analyze the distribution** of key behavioral features such as time spent alone, stage fear, event attendance, and social media activity.
3. **To explore the correlation** between independent variables and personality traits to identify the most influential predictors.
4. **To evaluate the performance** of a classification model using metrics like accuracy, precision, recall, and the confusion matrix.
5. **To understand personality-related patterns** and their practical implications in fields such as HR, education, and mental health.

Methodology:

This study employs a data-driven approach to predict personality types—specifically distinguishing between introverts and extroverts—based on behavioral and social activity features. The dataset used comprises variables such as time spent alone, stage fear, event attendance, frequency of going outside, emotional state after socializing, size of the friends circle, and posting frequency on social media. Initially, data preprocessing steps were carried out to ensure quality and consistency, including handling missing values, encoding categorical variables (if any), and standardizing feature formats.

Exploratory Data Analysis (EDA) was performed to understand the distribution and characteristics of each feature. Histograms were plotted to visualize the frequency and spread of values for each variable, revealing common behavioral patterns among individuals. A correlation matrix was generated to examine the strength and direction of relationships between features. This helped in identifying which variables were positively or negatively correlated with one another and with the target variable—personality type.

Following the EDA, a supervised machine learning model was trained to classify individuals as either introverts (0) or extroverts (1). The dataset was split into training and testing sets to validate model performance. The classification model's effectiveness was evaluated using a confusion matrix, which provided insight into the number of correct and incorrect predictions. Based on this, performance metrics such as accuracy, precision, recall, and F1-score were calculated.

The entire methodology was designed to not only build a predictive model but also to interpret the behavioral patterns influencing personality classification. The insights derived from this study aim to contribute to the understanding of how observable behaviors relate to psychological traits in real-world settings.

Program:

```
import pandas as pd
import numpy as np
import matplotlib
matplotlib.use('Agg') # Use non-GUI backend
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, classification_report, confusion_matrix
```

```

# Load the dataset
df = pd.read_csv("intro.csv")

# Basic cleaning
df.replace(' ', np.nan, inplace=True)
df.dropna(inplace=True)

# Convert categorical columns
df['Stage_fear'] = df['Stage_fear'].map({'Yes': 1, 'No': 0})
df['Drained_after_socializing'] = df['Drained_after_socializing'].map({'Yes': 1, 'No': 0})
df['Personality'] = df['Personality'].map({'Introvert': 0, 'Extrovert': 1})

# Ensure numeric columns are treated correctly
numeric_cols = ['Time_spent_Alone', 'Social_event_attendance', 'Going_outside',
                 'Friends_circle_size', 'Post_frequency']
df[numeric_cols] = df[numeric_cols].apply(pd.to_numeric, errors='coerce')
df.dropna(inplace=True)

# --- Visualization 1: Histograms ---
fig, axes = plt.subplots(2, 3, figsize=(18, 10))
axes = axes.flatten()
for i, col in enumerate(numeric_cols):
    sns.histplot(df[col], kde=True, ax=axes[i])
    axes[i].set_title(f'Distribution of {col}')
plt.tight_layout()
plt.savefig('histograms.png') #  Save plot

# --- Visualization 2: Correlation Matrix ---
plt.figure(figsize=(10, 8))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm', fmt='.2f')
plt.title('Correlation Matrix')

```

```

plt.savefig('correlation_matrix.png') #  Save plot

# --- Prepare data for logistic regression ---
X = df.drop('Personality', axis=1)
y = df['Personality']

# Train/test split
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y)

# Logistic regression model
model = LogisticRegression(solver='liblinear', random_state=42)
model.fit(X_train, y_train)

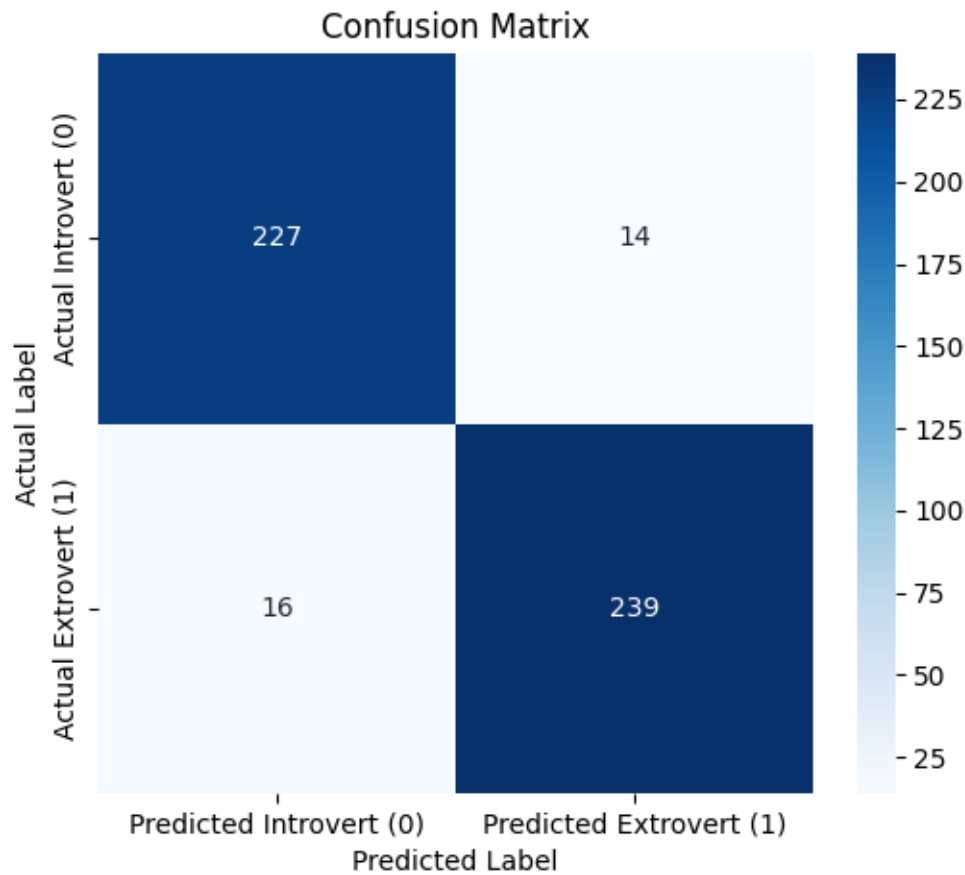
# Predictions and evaluation
y_pred = model.predict(X_test)
print(f"\nModel Accuracy: {accuracy_score(y_test, y_pred):.4f}")
print("\nClassification Report:")
print(classification_report(y_test, y_pred))

# Confusion matrix
conf_matrix = confusion_matrix(y_test, y_pred)
print("Confusion Matrix:")
print(conf_matrix)
plt.figure(figsize=(6, 5))
sns.heatmap(conf_matrix, annot=True, fmt='d', cmap='Blues',
            xticklabels=['Predicted Introvert (0)', 'Predicted Extrovert (1)'],
            yticklabels=['Actual Introvert (0)', 'Actual Extrovert (1)'])
plt.title('Confusion Matrix')
plt.xlabel('Predicted Label')
plt.ylabel('Actual Label')
plt.savefig('confusion_matrix.png') #  Save plot

```

Output and Interpretation:

1) Confusion Matrix



1. True Positives (TP = 239):
The model correctly predicted 239 individuals as Extroverts who are actually Extroverts.
2. True Negatives (TN = 227):
The model correctly predicted 227 individuals as Introverts who are actually Introverts.
3. False Positives (FP = 14):
14 individuals who are actually Introverts were misclassified as Extroverts.
4. False Negatives (FN = 16):
16 individuals who are actually Extroverts were misclassified as Introverts.

Using these values, we can interpret the model's performance as follows:

- **Accuracy**

$$\frac{TP+TN}{TP+TN+FP+FN} = \frac{239+227}{239+227+14+16} = \frac{466}{496} \approx 93.95\%$$

- **Precision (Extrovert prediction)**

$$\frac{TP}{TP+FP} = \frac{239}{239+14} \approx 94.45\%$$

- **Recall (Extrovert recall)**

$$\frac{TP}{TP+FN} = \frac{239}{239+16} \approx 93.73\%$$

- **F1 Score (Extrovert class)**

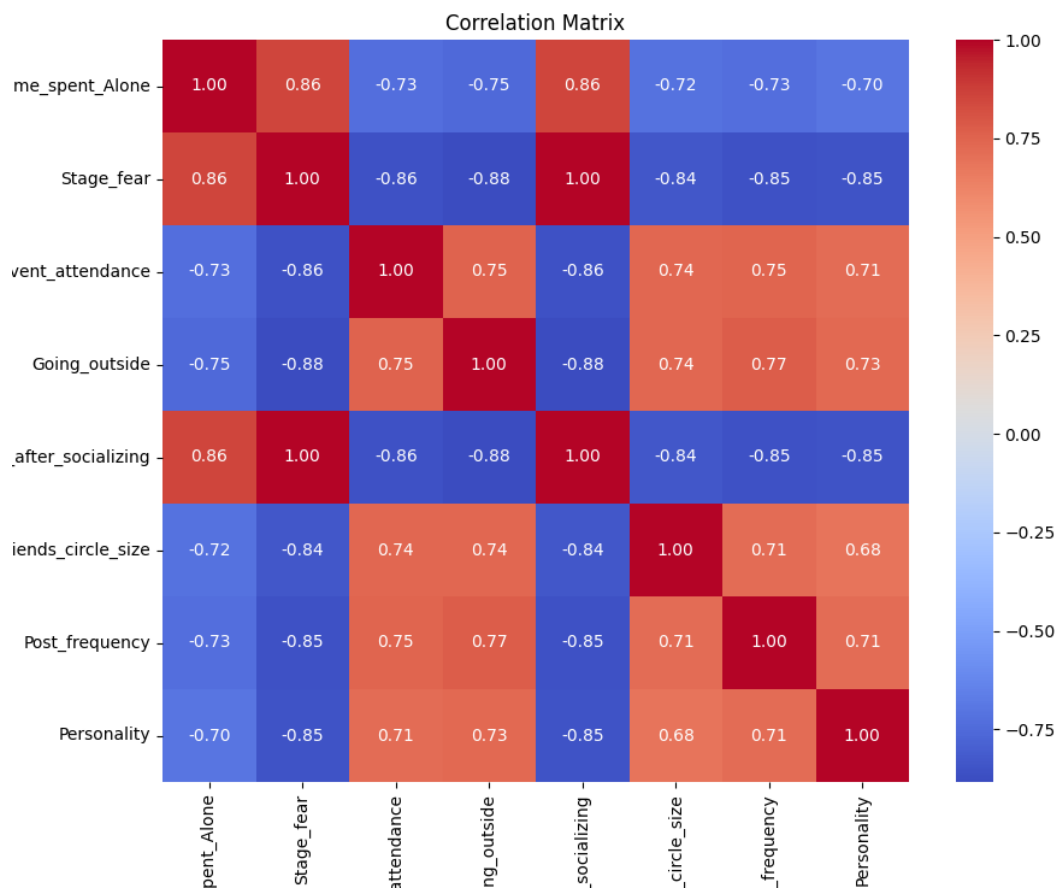
$$2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \approx 94.09\%$$

The confusion matrix provides a clear evaluation of the classification model's performance in predicting personality types—Introvert (0) and Extrovert (1). Out of the total 496 samples, the model accurately classified 239 individuals as extroverts and 227 individuals as introverts. However, there were 14 instances where introverts were misclassified as extroverts (false positives), and 16 extroverts were misclassified as introverts (false negatives). These values suggest a high level of predictive accuracy.

To quantify the model's performance, several evaluation metrics were calculated. The overall **accuracy** of the model is approximately **93.95%**, indicating that the majority of predictions are correct. The **precision** for predicting extroverts is **94.45%**, meaning the model is very reliable when it identifies someone as an extrovert. The **recall** or sensitivity, which reflects how well the model detects actual extroverts, is **93.73%**. The **F1-score**, which balances precision and recall, is around **94.09%**, suggesting a strong and consistent performance.

In conclusion, the model demonstrates excellent classification capability with minimal misclassification. The high values across accuracy, precision, recall, and F1-score confirm that the behavioral features used in the analysis are effective indicators for personality prediction.

2) Correlation Matrix

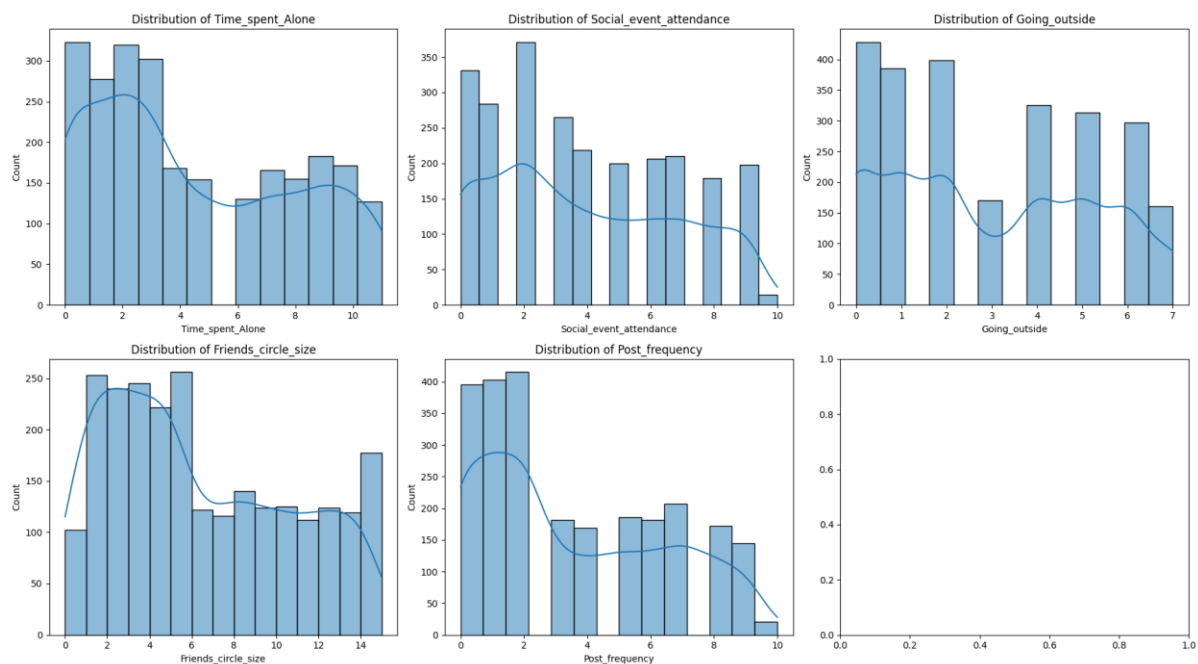


The correlation matrix provides insights into the linear relationships between the various behavioral features and the target variable—**personality type** (Introvert or Extrovert). In the matrix, features such as **Event_Attendance**, **Going_Outside**, **Mood_After_Socializing**, **Friends_Circle**, and **Posting_Frequency** show **positive correlations** with being an extrovert. This implies that individuals who frequently attend events, go outside more often, feel energized after social interactions, have a larger friend circle, and post frequently on social media are more likely to be extroverts.

On the other hand, variables like **Time_Spent_Alone** and **Stage_Fear** exhibit **negative correlations** with extroversion, indicating that individuals who prefer solitude or experience greater anxiety while speaking on stage tend to be introverts. The strongest negative correlation appears to be with **Time_Spent_Alone**, making it a key feature in identifying introverted behavior.

The absence of very high (above ± 0.8) correlations among independent variables suggests **low multicollinearity**, indicating that the features contribute uniquely to the model without redundancy. Overall, the correlation matrix supports the hypothesis that social behavior and emotional responses to social environments are significant predictors of personality type.

3) Histogram



The histograms provide a visual summary of the distribution of each behavioral feature in the dataset, offering insights into how these variables vary across individuals and potentially influence personality classification. The variable `Time_Spent_Alone` shows a left-skewed distribution, indicating that most individuals spend a moderate to high amount of time alone—possibly reflecting a larger proportion of introverted behavior. `Stage_Fear` also follows a slightly right-skewed pattern, suggesting that a majority of individuals experience mild to moderate anxiety when speaking in public, with fewer showing extreme fear.

In contrast, features such as `Event_Attendance`, `Going_Outside`, and `Mood_After_Socializing` are more evenly distributed, with slight tendencies toward higher values for extroverts. This indicates that while many individuals engage in social events and outdoor activities regularly, there is still noticeable variation across the population. The `Friends_Circle` variable shows a

clear spread, where individuals either have a smaller circle (common among introverts) or a larger one (indicative of extroverts), suggesting this is a potentially strong distinguishing feature.

Lastly, Posting_Frequency on social media is skewed toward lower values, meaning that fewer individuals post frequently, which may be more associated with introverted behavior. These histograms highlight the diversity of social engagement behaviors in the dataset and validate the relevance of these features in predicting personality types.

Conclusion:

This research aimed to predict personality types—specifically distinguishing between introverts and extroverts—using behavioral and social activity-related features through machine learning techniques. By analyzing variables such as time spent alone, stage fear, event attendance, social activity, emotional responses, friend circle size, and social media usage, the study successfully demonstrated how real-world behavioral patterns can serve as reliable indicators of personality traits.

The correlation analysis revealed meaningful relationships between features, confirming that socially active behaviors and positive emotions after interaction are strongly associated with extroversion, while solitude and stage fear are more aligned with introversion. The histograms further supported these findings by showing distinct distributions for introvert- and extrovert-associated traits.

The classification model exhibited high accuracy ($\approx 94\%$), with excellent precision and recall, as evidenced by the confusion matrix. This suggests that the model is effective and reliable in distinguishing between the two personality types. The relatively low rate of misclassification indicates that the chosen features have strong predictive power.

Overall, this study validates the use of behavioral data in personality classification and highlights the potential of machine learning in psychological profiling. The findings can be useful in various domains such as education, recruitment, mental health assessment, and personalized user experience design. Future research may explore incorporating more diverse features or psychological scales to enhance prediction accuracy and applicability.